Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

José Elías Durán Roa

Data Cluster · Directorate for Education and Skills



Why this project

- We study the decision to leave teaching. That needs each person's leave-or-stay outcome, plus covariates.
- Administrative records count best — but access is hard, coverage is partial, covariates are few.
- So we build the outcome from a household survey. This talk is about doing that carefully.

Contribution

1. **A user's guide** to building a leaving rate from a survey: each choice, each error, each effect on the number.

2. **An implementation** on 28 years of the CPS, checked against the NCES reference.

3. **The substance:** who leaves, and what correlates with it.

1

The data

Where do the official numbers come from?

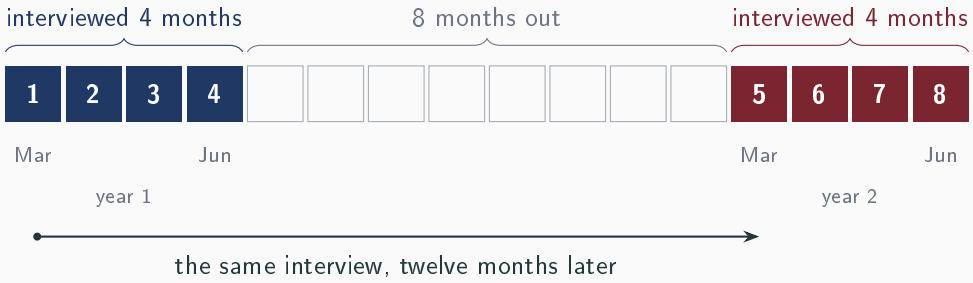
- National reference: the NCES Teacher Follow-up Survey. Itself a survey. Eight waves since 1988.
- Administrative data: state by state, public sector, and they measure system exit, not profession exit.
- No national administrative count exists.

The CPS replicates the official picture, every year

	CPS	NTPS	Difference
Measurement	**every year, 1997–2024**	occasional waves	
Example wave: 2020–21, the latest NTPS			
Female (%)	75.5	76.8	−1.3
Master's degree or higher (%)	57.0	61.0	−4.0***
Average age (years)	44.0	42.9	+1.1***
Teachers observed in the wave	3,855	39,633	

Public school teachers, weighted. Stars test the CPS mean against the NTPS estimate: *** $p < 0.001$.
Sources: NCES 2022-113, Digest 209.10, NCES 2024-024.

Full sample: **5.0 million** records, **104,545** teacher observations, **48,842** teachers in the linked panel.


interviewed 4 months
8 months out
interviewed 4 months
1
2
3
4
5
6
7
8
Mar
Jun
Mar
Jun
year 1
year 2
the same interview, twelve months later

Who counts as a teacher

Occupation of the longest job held last year, CPS ASEC — three Census classifications, 28 survey years

Surveys 1998–2002 teaching years 1997–2001 *1990 Census classification*		Surveys 2003–2019 teaching years 2002–2018 *2002/2010 Census classification (SOC)*		Surveys 2020–2025 teaching years 2019–2024 *2018 Census classification (SOC 2018)*	
155	Teachers, prekindergarten and kindergarten	2300	Preschool and kindergarten teachers	2300	Preschool and kindergarten teachers
156	Teachers, elementary school†	2310	Elementary and middle school teachers†	2310	Elementary and middle school teachers†
157	Teachers, secondary school†	2320	Secondary school teachers	2320	Secondary school teachers
158	Teachers, special education	2330	Special education teachers	2330	Special education teachers
159	Teachers, n.e.c. (not elsewhere classified)	2340	Other teachers and instructors	2360	Other teachers and instructors

Never included (they carry their own codes, outside the set): postsecondary teachers (2200s) · tutors (2350, 2018 cl.) teaching assistants (2540/2545) · childcare workers (4600) · education administrators (0230)

Are the codes traceable across census years?

Yes as a block: the K–12 teaching group maps one-to-one across the three classifications. Every figure uses the union of the codes, never a sub-code across regimes.

† sub-codes are not: middle-school teachers sit under elementary or secondary before 2003 and with elementary (2310) after; "other teachers" is renumbered 2340 → 2360 in 2020.

Yes within each survey: last year's job (OCCUP) and the current job (PEIOOCC) share one classification, so no leaver/switch comparison straddles a code change (seam surveys: 2003 and 2020).

Yes for people, but only two years: the CPS links the same person across two consecutive March interviews (PERIDNUM), never longer.

2

Measuring the leaving rate

Two ways to measure it

Follow the person

No person identifier in the CPS. We link on:

household identifier
dwelling identifier
person line number

checked: sex, race, age +0–2

Ask about last year

Each March, the supplement asks about the previous calendar year: longest job, weeks, earnings.

Taught last year, not teaching now $\Rightarrow$ left.

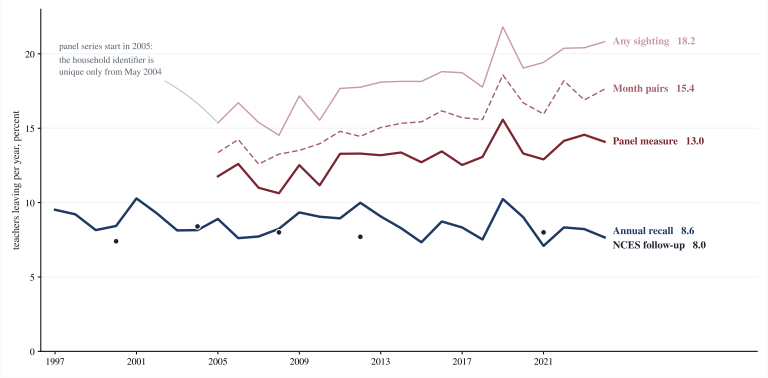
The measure used in the literature.

Four definitions, one language

Four interviews in year 1, four in year 2, twelve months apart.

	A teacher:	A leaver:
Any sighting	seen teaching in ≥ 1 of the 4 year-1 interviews	seen teaching in none of the 4 year-2 interviews
Month pairs	each teaching interview counts on its own	not seen teaching twelve months later
Panel (benchmark)	seen teaching in ≥ 2 of the 4 year-1 interviews	seen teaching in none of the 4 year-2 interviews*
Annual recall	teaching was the longest job of last year	not teaching at the March interview

One dataset, one rate per definition



Message. The definition moves the rate by a factor of four.

Policy. Fix the definition before comparing studies.

Recall: ≈3,700 teachers per year. Panel: ≈2,500. Dots: NCES follow-up.

Why the lines differ

18.2 Any sighting. Lets in substitutes and one-month spells.

15.4 Month pairs. Counts the same person several times.

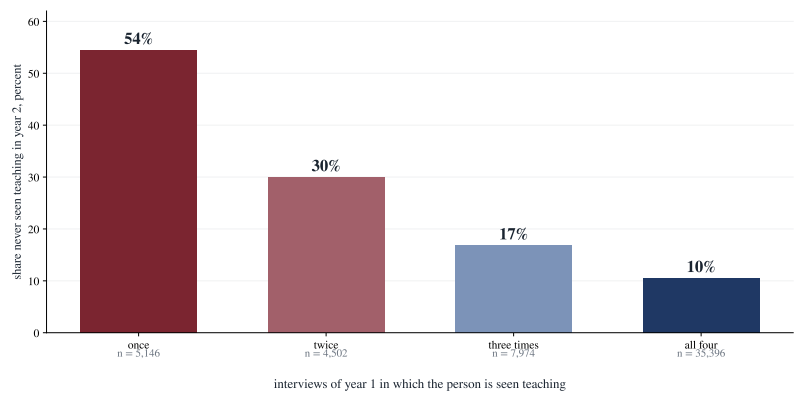
13.0 Panel. One verdict per person — but one leaver in six is back within three months.

8.6 Annual recall. The official concept: the year's main job.

8.0 NCES follow-up. The reference.

The choice, not the data, drives the spread.

Seen once, gone



Message. Seen teaching once: 54 percent not seen teaching a year later. Seen four times: 10.

Policy. Linkage follows addresses: movers drop out, and movers leave more.

3

What the data show

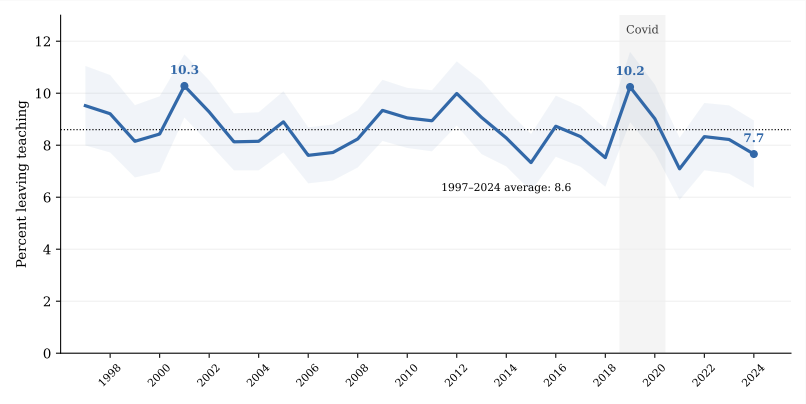
Why teachers leave: what the CPS sees

Reasons in the literature	In the CPS?
Working conditions	no
Organizational support	no
Autonomy and intrinsic motives	no
Financial factors	yes
Personal life-cycle motives	yes
Sociodemographics	yes

Five messages

1. Economics beats demographics.
2. Exits concentrate among part-time, part-year, low-paid teachers.
3. Those exits leave employment — not for better jobs.
4. A birth is the largest single trigger.
5. Pensions retain, and states differ widely.

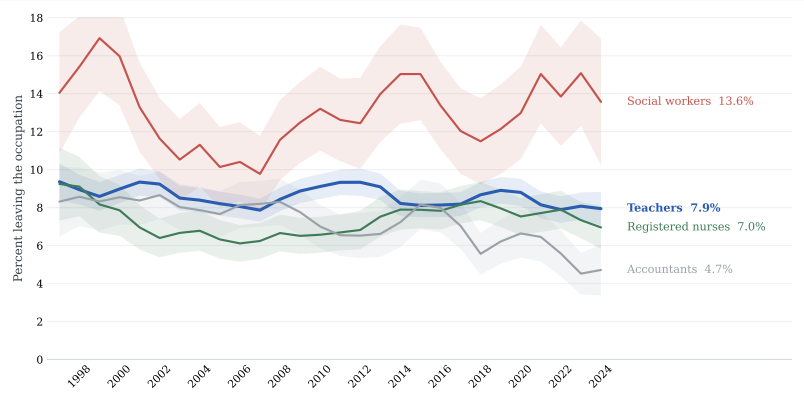
Leaving teaching, 1997–2024



Message. Flat around 8.6 since 1997.

Policy. A steady cost, not an emergency.

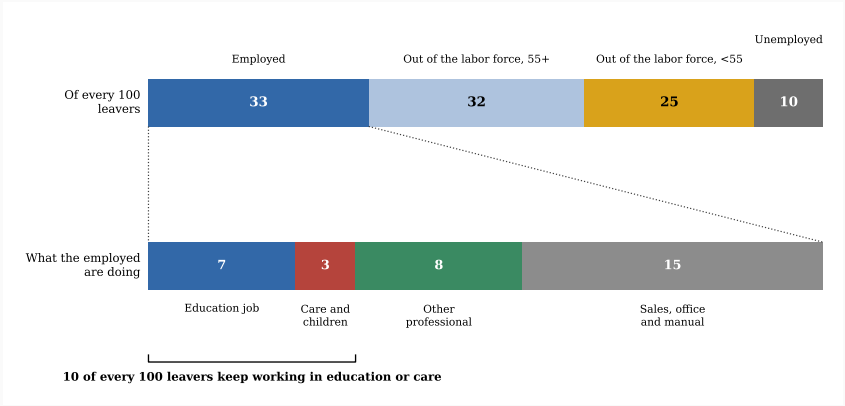
Teachers next to comparable professions



Message. Teachers are no outlier.

Policy. Benchmarks temper alarm.

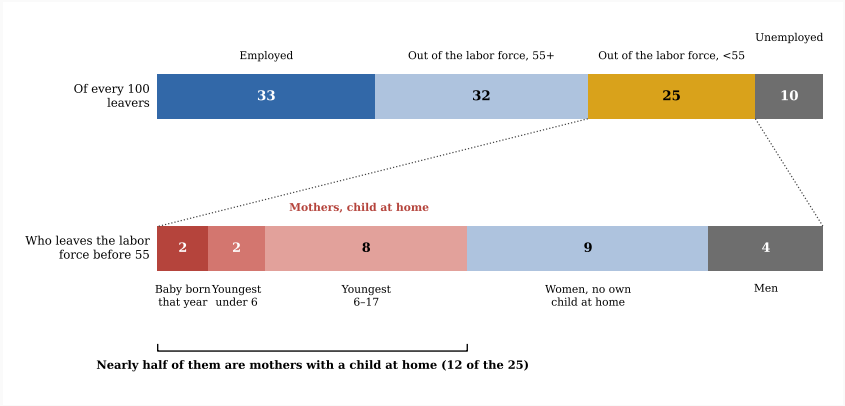
Destinations, 2024



Message. A third find another job; most leave employment.

Policy. The rival is retirement and care, not other employers.

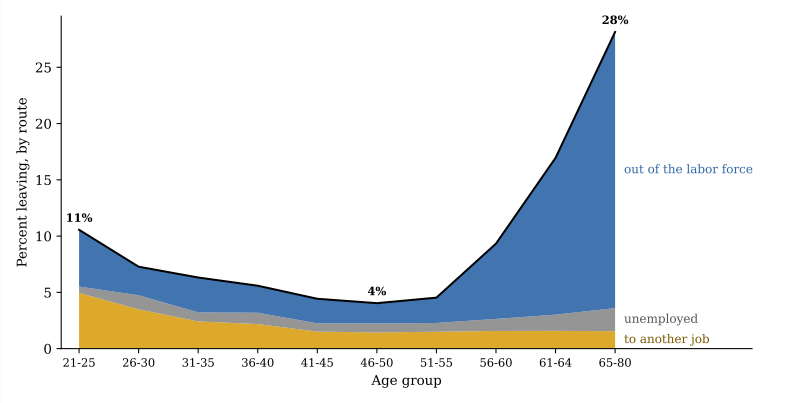
Who leaves the labor force before 55



Message. Half of the early exits are mothers.

Policy. Childcare policy is retention policy.

By age and route



Message. U-shaped in age; non-employment at both ends.

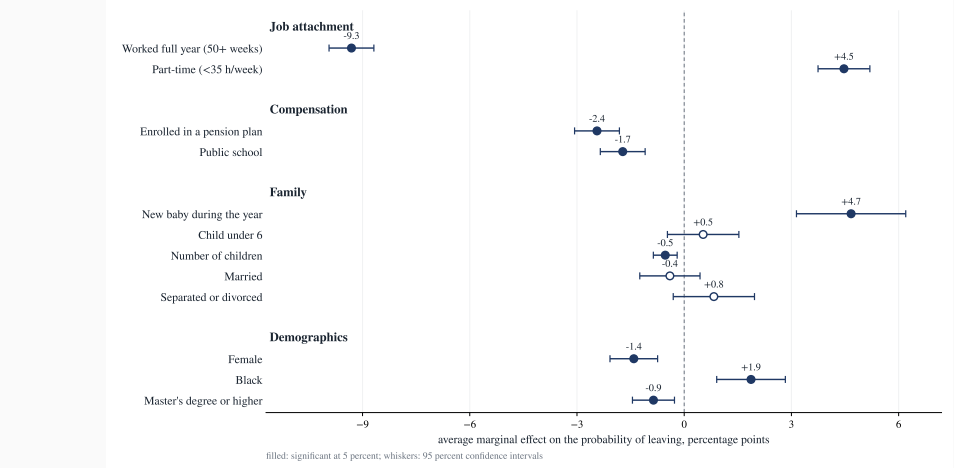
Policy. Different exits, different instruments.

The econometric specification

$$\Pr\big(\text{Leave}_{it} = 1 \mid x_{it}\big) \;=\; \Phi\Big(\underbrace{\text{attach}'_{it}\beta_1}_{\substack{\text{full year,}\\ \text{part-time}}} + \underbrace{\text{comp}'_{it}\beta_2}_{\substack{\text{pension,}\\ \text{public school}}} + \underbrace{\text{fam}'_{it}\beta_3}_{\substack{\text{new baby, children,}\\ \text{marital status}}} + \underbrace{\text{dem}'_{it}\beta_4}_{\substack{\text{age, age}^2\text{, sex,}\\ \text{race, education}}} + \delta_t\Big)$$

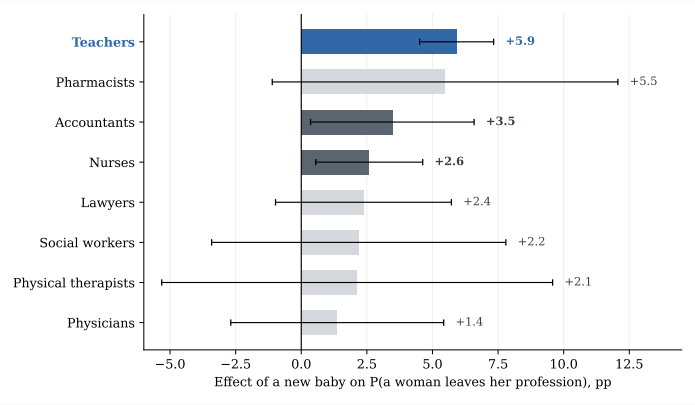
- Probit, 2015–2024, year effects, $n = 28{,}900$. AMEs in points.
- Robust: logit, LPM, state and year FE, full window, occupation-pair definition.

What correlates with leaving



Message. Attachment and compensation dominate.

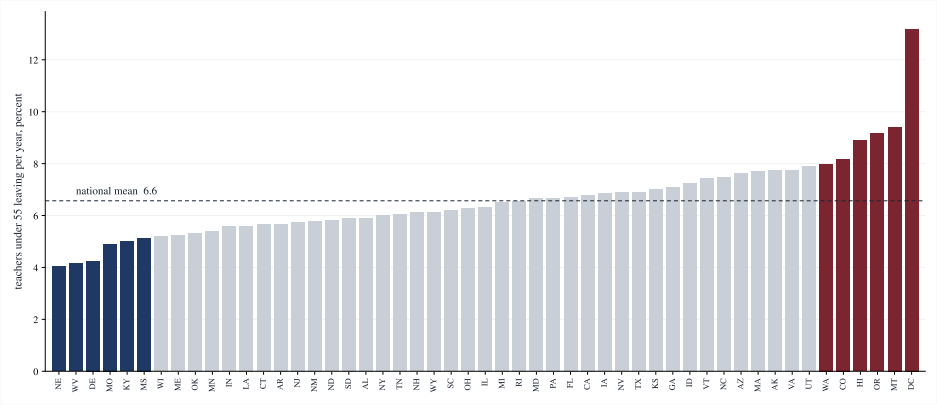
A new baby, by profession



Message. A birth adds 5.9 points — the most of any profession.

Policy. Parental leave targets the largest trigger.

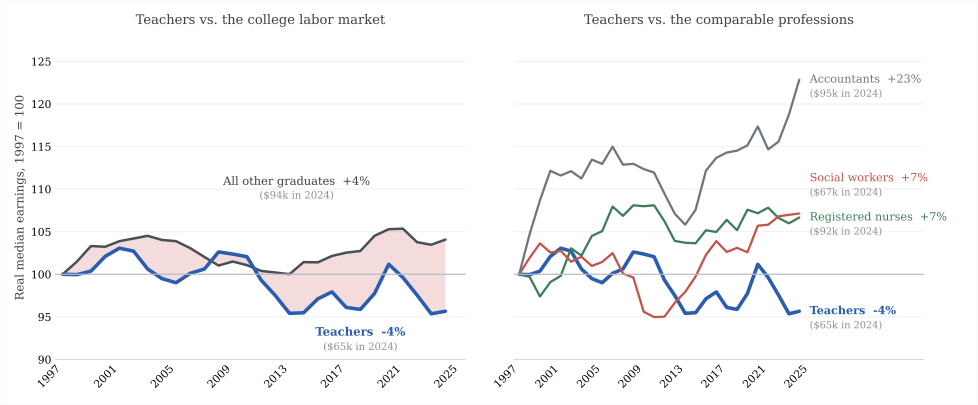
Across states



Message. From 4 to 13 percent across states.

Policy. State institutions are first-order.

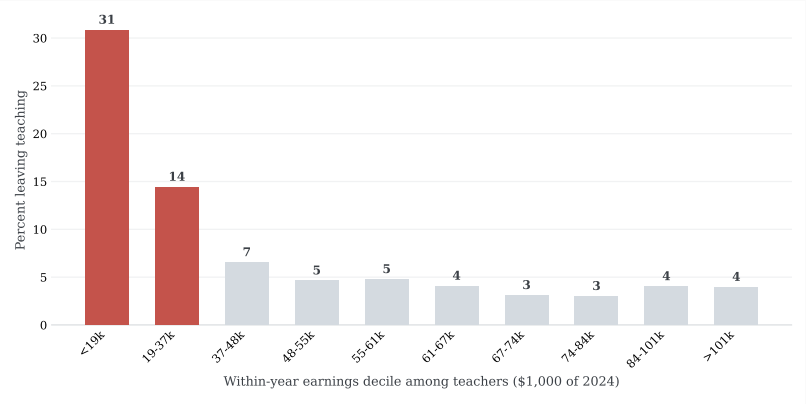
Relative pay since 1997



Message. Teacher pay fell 4 percent; other graduates gained.

Policy. A stock, not a shock.

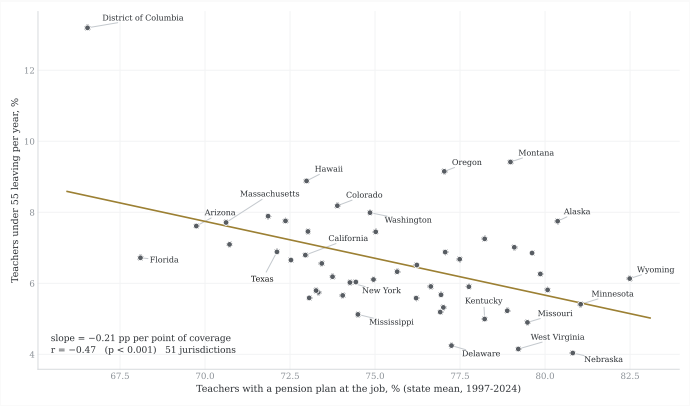
By position in the pay distribution



Message. Exits pile up in the lowest deciles.

Policy. Stabilize the bottom.

Pension coverage, across states



Message. More coverage, fewer leavers.

Policy. Pension design retains.

Conclusion

- A labour force survey measures attrition yearly — and matches the official picture.
- The definition is first-order: 5 to 19 percent from the same records.
- Attachment, pay, pensions and births drive exits. And the design travels to any LFS.